Genetic Conditions: Frame a Safe Research Question

LAUNCH · Science · 40 minutes

I can turn the SoW topic into a respectful research question, plan a school-approved source search, and separate biological explanation from personal or medical claims.

Prepare before pupils arrive

Print the pupil pack and open Lesson.pptx. Keep the uploaded HTML available for its live controls. Prepare any approved atlas/source extracts, existing learner work, authorised local data or prior planner named in the source lesson.

Prepare the approved local source set and privacy-safe access routes; keep the 1BI0 Foundation decision central.

Keep optional IGCSE depth staff-directed and separate from the core class route; do not use legacy 4SS0 framing.

Staff preview Keep the 1BI0 Foundation explanation, approved local source set, privacy boundary and learner-authored evidence decision central. Optional IGCSE depth remains staff-directed.

Staff qualification / truth boundary Core teaching anchor: Edexcel GCSE Biology 1BI0 Foundation / Combined Science 1SC0 Foundation. IGCSE 4BI1/4CH1/4PH1 is optional stretch for eligible pupils only. No diagnosis or medical advice. The SoW cell adds no unit, code, criterion or level. This lesson imports none of those as new criteria. Staff-only stretch: eligible pupils may use approved 4BI1/4CH1/4PH1 depth. It never governs the core class route. Legacy 4SS0 framing is not the governing route.

For review or assessment lessons, prepare the centre-approved questions, outcomes and existing learner records. Practice cards and a completed worksheet do not supply missing assessment evidence.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival · 3 min	Choose a private statement: I remember gene/allele links; I need a diagram; I need a full recap. No personal history is requested.	Offer private retrieval, pointing and a pass-and-return route; never request personal or family information. Change modality, density or number of choices before increasing prompting.

Stage / total time	Teaching move	Adult support / response
Starter · 3 min	Which question is safer and more scientific: one asking how evidence supports an inheritance explanation, or one asking pupils to disclose family information? Explain the boundary.	Receive a first decision before feedback; keep it visible for revision. Ask for the source or boundary clue, not confidence or personal experience.
I do · 4 min	Respectful research question checker Choose a draft question, then judge it. The condition is the one your teacher approved.	Model a neutral teacher-approved example; do not invent a learner biography, diagnosis or completed evidence. Name question, evidence, reasoning and limit as separate moves.
We do · 3 min	Select the checks that make a source useful for this question, not merely easy to read. Authority Who published the item, for what audience, and under what scientific or public-information responsibility?	Give private think time before receiving a spoken, pointed or written choice. Ask what evidence or boundary clue changes the claim.
I do 2 · 3 min	Use neutral language: a person with a condition, unless a teacher-approved source explains a community preference. Separate inheritance probability from certainty about any individual person.	Keep diagnosis, advice, personal data and unsupported certainty outside the model. Model a repair or an honest hold when the source set cannot answer.
We do 2 · 4 min	1 · Locate Write a bounded question and identify two teacher-approved source cards or locally saved extracts. 2 · Connect Match each source type to one science field: inherited biology, probability, variation, evidence or uncertainty.	Let the learner operate or direct the shared source map. Use the explanation to select the lightest independent scaffold.
Independent · 16 min	Create a research brief for one teacher-approved genetic condition using no personal or family data. Record the question, keywords, two approved source types, the evidence categories required, a respectful-language check and one uncertainty that must remain visible.	Use least-prompt-first and keep exact-word scribing visibly adult-supported. Capture only genuine learner decisions and meaning; the HTML stores nothing.
Exit · 4 min	Submit the question and source plan privately. State one claim the research may support and one claim it must not make.	Receive the learner's evidence and limit; do not upgrade either. Name the receiving role and the next report-back point.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

How do approved sources explain the inheritance pattern and its uncertainty?

This is bounded: it asks what approved evidence can explain and does not ask about a real pupil or family.

Who in our class or families may have this condition?

This leaves the C44 boundary because it asks for personal and family information. Replace it with a source-answerable question.

Can the presentation predict exactly what will happen to one person?

This overclaims. Research evidence can describe inheritance and variation; it cannot guarantee an individual's future.

1 • publisher and date

Useful: a named publisher and date create a source trail that another person can locate.

2 • first search result

Not enough: popularity does not establish scientific purpose, evidence or currency.

3 • purpose and evidence

Useful when bounded: purpose, evidence type and limits help decide what the source can support.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre’s authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner’s practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_L_W14L1_Genetic_Condition_Research_Introduce.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.